ORIGINAL ARTICLE

Risk for hepatitis B and C virus reactivation in patients with psoriasis on biologic therapies: A retrospective cohort study and systematic review of the literature

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Background: Patients with psoriasis on biologic therapies and a history of viral hepatitis carry a risk for reactivation.

Objective: We evaluated safety of biologic therapies in psoriasis patients seropositive for hepatitis B or C viruses (HBV, HCV).

Methods: A retrospective cohort study design was used. Clinical and laboratory data for 30 patients undergoing biologic therapy who were seropositive for HBV or HCV were evaluated. Next, a systematic review was performed. Primary outcomes were hepatitis and viral reactivation during therapy. Treatment duration and antiviral prophylaxis were also recorded.

Results: Serology indicated HCV infection in 4 patients, past HBV infection in 17 patients, isolated core antibody in 8 patients, and chronic HBV infection in 1 patient. During follow-up (mean 4.85 ± 3.1 years), no patients experienced hepatitis or viral reactivation. The systematic review of the literature included 49 studies comprising 312 patients followed for a mean of 30.9 months. Viral reactivation occurred in 2/175 patients who were seropositive for core antibody and 3/97 with HCV infection (yearly rates, 0.32% and 2.42%, respectively) compared with 8/40 patients with chronic HBV infection (yearly rate, 13.92%). Three of these 8 patients with reactivated HBV infection received antiviral prophylaxis.

Limitations: We pooled heterogeneous studies evaluating different biologic therapies.

Conclusion: Biologic therapies pose minimal risk for viral reactivation in low-risk patients without hepatitis seropositive for HCV or HBV core antibody but are a considerable risk in patients with chronic HBV infection, highlighting the necessity of antiviral prophylaxis. (J Am Acad Dermatol http://dx.doi.org/10.1016/j.jaad.2017.01.037.)

Key words: biologic; HBcAg; hepatitis B virus; hepatitis C virus; psoriasis; reactivation; risk.

In the last 15 years, biologic therapies have been introduced for the treatment of moderate-to-severe psoriasis.[1] The main therapies available in Israel are anti—tumor necrosis factor-α (TNF-α), the interleukin (IL) 12/23 p40 monoclonal antibody ustekinumab, and the recently approved IL-17 antibody secukinumab.[2] Because these agents lack direct hepatic or renal toxicity,[3] these selective immunomodulatory treatments are appealing to patients with concurrent viral hepatitis.

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With the advent of TNF-α—inhibitor use in patients, so did reports on various disseminated and opportunistic infections.[4,5] TNF-α is an important cytokine in host antiviral defense.[6] Reports describing hepatitis B virus (HBV) reactivation following TNF-α—inhibitor use in patients with rheumatoid arthritis and spondyloarthropathies[7] prompted concerns regarding the safety of all biologics in psoriasis patients exposed to HBV or hepatitis C virus (HCV), but no systematic study addressing this issue has been conducted to date.

The aim of the present study was to evaluate the safety profile of biologic therapies among psoriasis patients with concurrent positive serology for HBV or HCV.

CAPSULE SUMMARY

- Patients with a history of viral hepatitis carry a reactivation risk, but it is uncertain how this is affected by biologic therapies for psoriasis.
- Patients positive for hepatitis C virus or hepatitis B virus (HBV) core antibody without evidence of hepatitis had a minimal risk for reactivation, whereas those with chronic HBV had a higher risk for reactivation.
- Select patients with a history of viral hepatitis may be safely treated with biologics.

METHODS

Retrospective cohort study

A retrospective cohort study design was used. The electronic database of the Dermatology Department of Rabin Medical Center, a tertiary referral hospital in Israel, was searched for patients treated with biologic agents for moderate-to-severe psoriasis since 2005. All patients were screened for HBV and HCV before initiation of biologic therapy. Positive HBV serology was subdivided into 3 categories according to the European Association for the Study of Liver Disease[8]: 1) resolved past HBV infection (HBV surface antigen [HBsAg] negative, HBV surface-specific antibody [HBsAb], and HBV core-specific antibody [HBcAb] positive), 2) chronic HBV infection (HBsAg positive), and 3) only core antibody positive (HBsAg negative, HBsAb negative, and HBcAb positive). Hepatitis was defined as an increase in the alanine transaminase level to 5 times the upper limit of normal.[9] Because no uniform diagnostic criteria are available for HBV reactivation, we adopted the definition commonly used by other researchers[10]: an increase in HBV replication of at least 1 log 10 copies/mL or conversion of serum HBV DNA results from negative to positive. HCV reactivation was defined as an increase of at least 1 log 10 copies/mL. According to Israeli guidelines, patients in whom pretreatment serology was suggestive of HBV or HCV infection were scheduled for clinical evaluation, which included measurement of transaminases, every 2-3 months until the end of follow-up. Patients with chronic HBV infection were serologically evaluated every 3-6 months. Because the best strategy for monitoring low-risk, only core antibody—positive patients is unknown,[11] the serologic status (HBsAg, HBcAb, and HBV DNA) of patients was evaluated at various intervals. The presence of HCV RNA was assessed at baseline and reassessed as part of antiviral therapeutic protocols, as suggested by the hepatologist. The demographic, clinical, and laboratory data were extracted onto a chart, which included the fields shown in Tables I-IV. The study was approved by the local ethics committee.

Systematic literature review

The systematic review was conducted and reported in accordance with the PRISMA (Preferred Reporting Items for Systemic Reviews and Meta-Analyses) statement.[12] We performed a search with no date limits using PubMed, our institute's search engine, and reference lists of included articles on May 13, 2016. The search criteria are shown in Appendix I (available at http://www.jaad.org).

Inclusion and exclusion criteria. We included articles written in English and excluded articles written in other languages. We selected articles describing the treatment of psoriasis with at least 1 biologic agent in the presence of a positive serologic or virologic marker for HBV or HCV before treatment onset that also had available serologic or virologic data on HBV or HCV at the end of follow-up. Any relevant study, regardless of design, was included. Literature reviews, expert opinions, and studies in which data of interest could not be extracted were excluded.

Data extraction and variables. One reviewer (IS) screened the titles and abstracts of all retrieved articles followed by the full text of the articles considered potentially eligible and extracted the data into a predefined form (Supplemental Table I; available at http://www.jaad.org)[13-60] that included the following items: 1) HCV and HBV infection status at the beginning of the study, 2) treatment factors, such as treatment duration, types, and quantities of different biologics, and 3) evidence of viral reactivation or hepatitis.

Studies were graded according to their level of evidence: (A) prospective controlled study, (B)

Abbreviations used:

HBcAb:	HBV core-specific antibody
HBsAb:	HBV surface-specific antibody
HBsAg:	HBV surface antigen
HBV:	hepatitis B virus
HCV:	hepatitis C virus
IL:	interleukin
OBI:	occult HBV infection
PRISMA:	Preferred Reporting Items for Systemic Reviews and Meta-Analyses
TNF-α:	tumor necrosis factor-α

retrospective study, or (C) case report or case series.[61]

Data analysis

Patients were divided into 3 groups: those with HCV infection; those with chronic HBV infection, defined as serologically positive for surface antigen; and those positive for the HBV core antibody, irrespective of surface antibody status. To calculate the viral reactivation, we included only patients with available HBV DNA levels at the end of follow-up.

Data were analyzed with descriptive statistics. Categorical variables are described with numbers and percentages, and quantitative variables are described with means and standard deviations or ranges.

RESULTS

Retrospective cohort study

Patients. Of the 207 patients with psoriasis treated with biologics during the study period, 30 (14.49%) had positive markers of HBV (n = 26) or HCV (n = 4) before treatment initiation.

Hepatitis B virus infection. The 26 patients with serologic evidence of HBV infection included 18 men and 8 women of mean age 56.3 (range 34-80) years. Their clinical (Table I) and laboratory (Table II) data are shown. All had long-standing plaque psoriasis (average duration 14.6 years, range 0.5-40 years); 2 patients (cases 7 and 14) also had a phase of erythroderma during disease exacerbation. Twenty patients previously had used ≥1 nonbiologic agent including methotrexate or cyclosporine. Etanercept was the most common biologic agent used (n = 24), followed by adalimumab (n = 15), ustekinumab (n = 11), infliximab (n = 4), secukinumab (n = 3), golimumab (n = 1), and alefacept (n = 1 each). Fifteen patients received >1 biologic agent (average 2.27, range 1-6). The mean duration of treatment including clinical follow-up was 5.31 (range 1-11) years.

Seventeen patients had serologic findings suggestive of a resolved past infection (HBsAg negative, HBsAb positive, HBcAb positive); 8 patients were seropositive for only core antibody (HBsAg negative, HBsAb negative, and HBcAb positive), and 1 patient was an inactive HBV carrier (HBsAg positive, HBV e antigen negative, and serum HBV DNA 122 IU/mL). No other patient had detectable serum HBV DNA at baseline or concurrent HCV infection. Two patients (cases 14 and 23) received antiviral prophylaxis with lamivudine 100 mg daily during treatment.

During follow-up, significant changes in serum liver enzyme activity, clinical signs of hepatitis, and liver failure were not found. Data on HBV viral load and serology at the end of follow-up were available for 25 patients (mean follow-up time 4.45 years, range 1-10.3 years). No patients had evidence of viral reactivation or HBsAg seroconversion.

A Psoriasis Area Severity Index 50 improvement from baseline was achieved in 18 patients (69%) at 3 months from the start of treatment. Several adverse effects were recorded during follow-up: case 3 had pneumonia, case 5 had 2 episodes of upper respiratory infection, and case 23 had a myocardial infarction.

Hepatitis C virus infection. The patients with serologic evidence of HCV infection were 4 men of mean age 62.3 (range 38-78) years. Their clinical (Table III) and laboratory (Table IV) data are shown. All had longstanding plaque psoriasis (average duration 32.3 years, range 15-60 years). One patient (case 1) had used methotrexate and cyclosporine before biologic therapy. All received treatment with etanercept, and one (case 1) was also treated sequentially with adalimumab. Mean duration of treatment and clinical follow-up was 1.86 (range 0.67-3) years. Data on the viral load at the end of follow-up were available for 3 patients (mean follow-up time 1.47 years, range 0.67-2.17 years). The viral isotype was available for 2 patients: 1 had genotype 2a2c and the other had genotype 1b. No patients received antiviral treatment. Serology in one patient (case 4) was also suggestive of past HBV infection, with borderline positivity for core antibody and positivity for HBsAb.

No patients exhibited any substantial change in hepatic enzymes or evidence of viral reactivation. Three patients achieved Psoriasis Area Severity Index 50 at 3 months, and no patients had severe adverse effects caused by treatment.

Systematic literature review and analysis

Included studies. Our search yielded 362 citations. Forty-eight already published studies and our data fulfilled the inclusion criteria (Fig 1) for a total of 49 studies. From our cohort study, we included only the 28 patients with available HBV DNA levels at the

Table I. Clinical characteristics of patients with HBV infection at baseline

Case	Sex/age	Medical history	Psoriasis duration, years*	HBV status	Previous immuno-therapy	Biologic therapies (n)	Treatment duration, years	PASI/BSA baseline	PASI 50 response
1	M/55	DM, Dys, HTN	0.5	Isolated	—	ADA (1)	1	PASI 26.1	Yes
2	M/44	Dys	20	Past	MTX, CsA	ETA, ADA, UST (3)	6	BSA 50	Yes
3	M/69	Dys, HTN	19	Isolated	—	ADA (1)	6	BSA 50	Yes
4	M/66	DM, Dys, HTN	20	Isolated	MTX	ETA, ADA, GOL, UST (4)	8	BSA 50	Yes
5	F/73	DM	5	Past	MTX	ETA (1)	5	BSA 50	Yes
6	M/57	Dys	N/A	Isolated	MTX	ETA (1)	9	BSA 50	Yes
7	M/47	—	10	Past	MTX, CsA	ALE, INF, UST, ETA, ADA, SEC (6)	11	PASI 22	No
8	M/54	DM, Dys	3	Past	MTX	ETA (1)	10	BSA 50	Yes
9	F/57	HTN	35	Past	MTX	ETA, ADA, SEC (3)	6	BSA 50	No
10	F/66	HTN	8	Past	MTX	UST, ETA, ADA (3)	6	PASI 13.2	Yes
11	F/52	DM, Dys, HTN	5	Past	—	ETA, ADA, UST (3)	4.5	PASI 22.2	Yes
12	M/80	—	40	Past	MTX	ETA, ADA (2)	4	PASI 20.1	Yes
13	F/63	DM, Dys, HTN	20	Past	—	ETA (1)	8	BSA 50	Yes
14	F/61	Dys, HTN	30	Past	MTX	ETA, ADA, UST, INF (4)	8	BSA 50	No
15	M/55	DM, HTN	6	Isolated	MTX	ETA, ADA, INF, UST (4)	10	BSA 50	Yes
16	M/43	Dys	1	Past	—	ETA, ADA, UST (3)	3.5	PASI 14.4	No
17	M/58	Dys	10	Isolated	MTX	ETA, UST, SEC (3)	2	PASI 27	No
18	M/65	HTN	32	Past	MTX	ETA, ADA (2)	3	PASI 19.2	Yes
19	M/43	DM, Dys	19	Past	CsA	ETA, UST (2)	3	N/A	N/A
20	F/53	Dys, HTN	13	Isolated	MTX	ETA (1)	3	BSA 70	Yes
21	M/67	Dys, HTN	N/A	Past	MTX	ETA (1)	2	PASI 22.6	Yes
22	M/34	Dys, HTN	16	Past	MTX	ETA (1)	1	PASI 21.6	Yes
23	F/62	—	2	Carrier	—	ETA (1)	1	PASI 18	Yes
24	M/48	—	6	Past	MTX	ADA, ETA (2)	1.17	PASI 17	N/A
25	M/34	Dys	N/A	Isolated	MTX, CsA	ETA, ADA, INF, UST (4)	8	PASI 40	N/A
26	M/59	Dys, HTN	N/A	Past	MTX, CsA	ETA (1)	8	PASI 32	Yes

ADA, Adalimumab; *ALE*, alefacept; *BSA*, body surface area; *CsA*, cyclosporine; *DM*, diabetes mellitus; *Dys*, dyslipidemia; *ETA*, etanercept; *GOL*, golimumab; *HBV*, hepatitis B virus, *HTN*, hypertension; *INF*, infliximab; *MTX*, methotrexate; *N/A*, not available; *PASI*, Psoriasis Area and Severity Index; *SEC*, secukinumab; *UST*, ustekinumab.
*All patients had plaque-type psoriasis; patient 7 and 14 had an episode of erythroderma.

end of follow-up. One study was prospective (grade A, 2 patients); 15 were retrospective (grade B, 265 patients), and 33 were either small case series or case reports (grade C, 45 patients). Sample sizes ranged from 1 to 62, with a total of 803.48 reported patient-years: HCV infected patients accounted for 123.83 patient-years, chronic HBV infected patients accounted for 57.46 patient-years, and patients positive for HBcAb accounted for 622.19 patient-years. The complete list of included studies appears in Appendix I (Supplemental Table I; available at http://www.jaad.org).[13-60] Table V[13-60] summarizes the main characteristics of the included studies, and Table VI summarizes the reports describing viral reactivation during biologic therapy.

Chronic HBV infection. In 40 patients, serology was compatible with chronic HBV infection. Mean follow-up time was 17.24 (range 0.75-39) months. On average, patients received 1.05 (range 1-2) different biologics. Viral reactivation was documented in 8 (20%) patients, and concurrent hepatitis in 1 patient, for a calculated yearly reactivation rate of 13.92%. Of the 26 patients with chronic HBV who received antiviral prophylaxis, 3 had viral reactivation, for an adjusted yearly reactivation rate of 7.74%, compared with 5/14 patients who did not receive prophylaxis, for an adjusted yearly reactivation rate of 26.31% (Supplemental Table II; available at http://www.jaad.org).

Resolved past HBV infection or core antibody only. Serology was positive for core antibody in 175 patients. Mean follow-up time was 42.66 (range 7.3-66) months. On average, patients received 1.4 (range 1-2.32) different biologics. Two patients received antiviral prophylaxis. Two patients (1.14%) had evidence of viral reactivation, for a yearly reactivation rate of 0.32%. None of the patients had hepatitis.

HCV infection. Serology was suggestive of HCV infection in 97 patients (Table V).[13-60] Mean follow-up time was 15.32 (range 3-46) months. On average, patients received 1.07 (range 1-3) different biologics.

Table II. Laboratory characteristics of patients with HBV infection

Case	HBsAg	HBsAb (mIU/mL)		HBcAb	Viral load (IU/mL)		AST (IU/L)		ALT (IU/L)		Antiviral PRX
		Baseline	Follow-up		Baseline	Follow-up (months)	Baseline	End	Baseline	End	
1	Negative	50.7	195	Positive	N/A	<20 (12)	35	29	34	42	No
2	Negative	2.22	78	Positive	N/A	0 (64)	22	34	26	28	No
3	Negative	4.31	N/A	Positive	N/A	0 (19)	19	17	24	14	No
4	Negative	N/A	N/A	Positive	N/A	0 (68)	24	22	18	13	No
5	Negative	591	321	Positive	N/A	<20 (60)	18	21	14	11	No
6	Negative	N/A	4.18	Positive	N/A	0 (84)	22	22	27	19	No
7	Negative	68.77	66.98	Positive	0	0 (124)	32	16	30	14	No
8	Negative	78.4	61.42	Positive	N/A	0 (83)	21	21	33	26	No
9	Negative	>1000	>1000	Positive	0	<20 (72)	20	27	12	23	No
10	Negative	407	169	Positive	<60	<20 (72)	23	28	24	28	No
11	Negative	21.7	8.72	Positive	<60	<20 (54)	14	22	15	20	No
12	Negative	11.3	22.45	Positive	<60	<20 (48)	N	N	N	N	No
13	Negative	N/A	>1000	Positive	N/A	<20 (79)	23	N	21	N	No
14	Negative	33.4	7.7	Positive	<20	<20 (39)	23	22	17	6	Yes
15	Negative	6.73	5.15	Positive	<20	<20 (116)	34	24	28	20	No
16	Negative	>1000	N/A	Positive	<20	<20 (42)	28	34	30	43	No
17	Negative	5.19	5.57	Positive	<20	<20 (29)	26	26	28	30	No
18	Negative	>1000	>1000	Positive	<20	<20 (36)	19	N	17	N	No
19	Negative	29.19	23.47	Positive	<20	<20 (29)	21	19	23	23	No
20	N/A	N/A	N/A	Positive	N/A	<20 (36)	20	22	18	17	No
21	Negative	35.92	N/A	Positive	<20	<20 (25)	67	36	36	33	No
22	Negative	548	561	Positive	N/A	<20 (22)	18	16	16	15	No
23	Positive*	1.54	N/A	Positive	122	<20 (12)	17	17	13	13	Yes
24	Negative	>1000	>1000	Positive	<20	<20 (14)	35	N	32	N	No
25	Negative	0	0	Positive	<60	<20 (96)	19	16	8	14	No
26	Negative	219	N/A	Positive	<60	N/A	15	17	18	10	No

There were no serious adverse effects of biologic therapies.
ALT, Alanine aminotransferase; *AST*, aspartate aminotransferase; *HBcAb*, HBV core antibody; *HBsAb*, HBV surface antibody; *HBV*, hepatitis B virus; *N*, within normal range; *N/A*, not available; *PRX*, prophylaxis.
*Patient was seronegative for HBV e antigen.

Table III. Clinical characteristics of patients with HCV infection at baseline

Case	Sex/age, year	Medical history	Psoriasis duration (years)*	Previous immuno-therapy	Biologic therapies (n)	Treatment duration, years	PASI/BSA baseline	PASI 50 response
1	M/78	Dys, HTN	60	MTX, CsA	ETA, ADA (2)	3	BSA 50	Yes
2	M/64	Dys, HTN, NMSC	22	—	ETA (1)	2.17	PASI 27	Yes
3	M/38	—	15	—	ETA (1)	1.58	PASI 17.4	Yes
4†	M/69	—	N/A	—	ETA (1)	0.67	PASI 14	N/A

ADA, Adalimumab; *BSA*, body surface area; *CsA*, cyclosporine; *Dys*, dyslipidemia; *ETA*, etanercept; *HTN*, hypertension; *HCV*, hepatitis C virus; *MTX*, methotrexate; *N/A*, not available; *NMSC*, nonmelanoma skin cancer; *PASI*, Psoriasis Area and Severity Index.
*All patients had plaque-type psoriasis.
†In addition to HCV, patient also had borderline positivity for core antibody and positivity for the hepatitis B virus surface antibody.

Evidence of viral reactivation was found in 3 patients (3.09%), for a yearly reactivation rate of 2.42% (Table VI). None of the 3 patients had evidence of hepatitis or received antiviral prophylaxis.

DISCUSSION

Up to 37.2% of patients with moderate-to-severe psoriasis are treated with biologic therapies.[63] Therefore, it is crucial to define the safety profile of these agents in patients with concurrent serology suggestive of viral hepatitis.

Our cohort included low-risk patients without evidence of hepatitis who were seropositive for HBV or HCV. None of these patients experienced hepatitis or viral reactivation, supporting the safety of using biologics in this setting. The findings are reinforced

Table IV. Laboratory characteristics of patients with HCV infection

		Viral load (copies/mL)			AST (IU/L)		ALT (IU/L)		
Case	Anti-HCV	Baseline	Follow-up (months)	Viral isotype	Baseline	End	Baseline	End	Anti viral PRX
1	Positive	4,281,177	N/A	N/A	68	32	74	32	No
2	Positive	25,020	64,856 (26)	2a2c	43	75	54	98	No
3	Positive	854,720	1,085,470 (19)	N/A	45	45	65	65	No
4	Positive	656,522	Not detected (8)	1b	57	41	57	54	No

ALT, Alanine aminotransferase; *AST*, aspartate aminotransferase; *HCV*, hepatitis C virus; *N/A*, not available; *PRX*, prophylaxis.

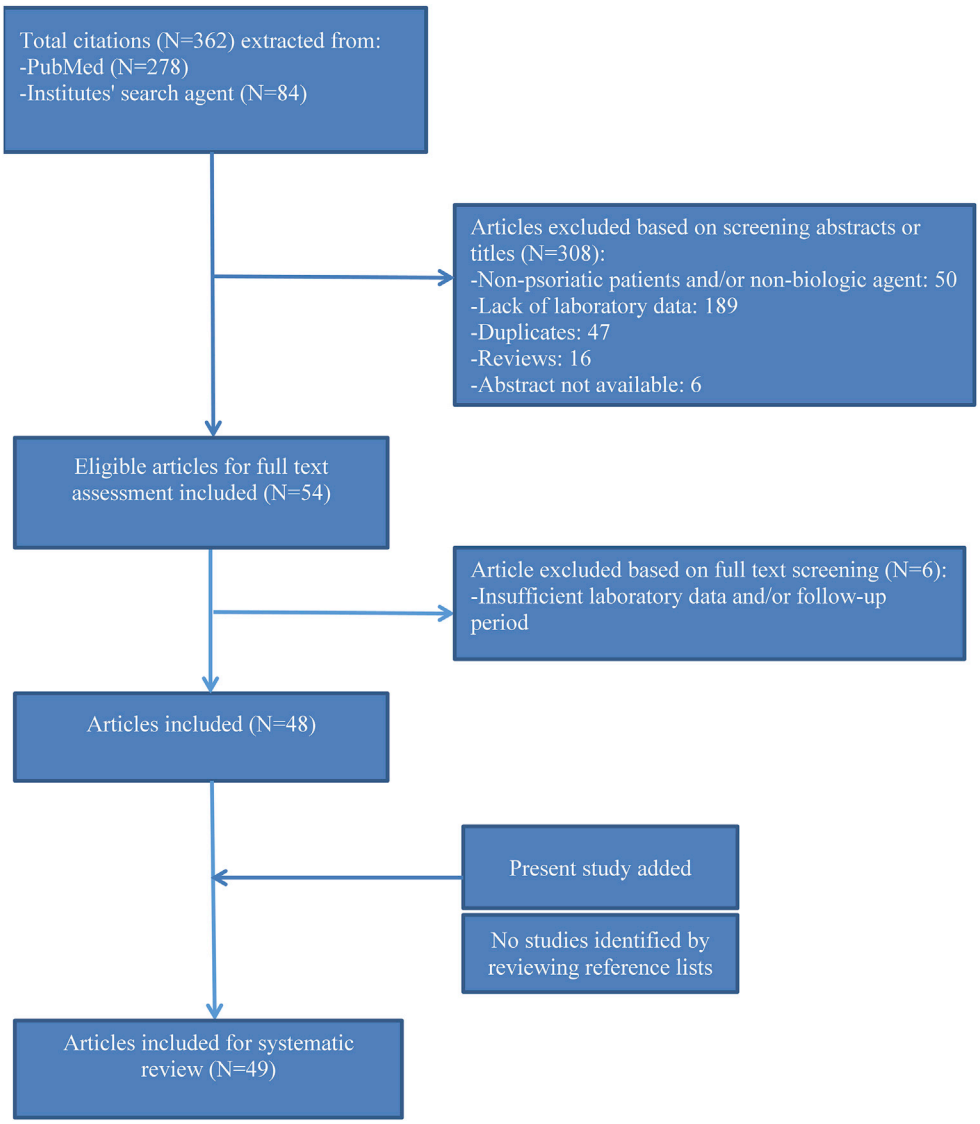

Fig 1. Flow chart showing study selection for systemic review.

by the strengths of our study: longer mean follow-up period (5.31 for HBV; 1.86 for HCV; 4.85 years overall) than in any previous cohort study; larger average number of biologic agents (2.13) than calculated in our systematic review (1.24); and first safety assessment of the recently approved IL-17 antibody, secukinumab, used by 3 patients with positive markers for HBV.

All patients with a history of exposure to HBV might be at risk for viral reactivation during immunosuppressive states.[64-69] Core antibody only positivity is a clinically distinct entity from occult

Table V. Summary of laboratory findings in studies included in the systematic review[13-60]

Viral profile	Pts, n	Follow-up time mean ± SD, years	Combined follow-up time, years	Reactivation rate, % (n/total)	Viral PRX rate,% (n/total)	Reactivation rate, %/year	Number of biologic therapies, mean (range)	Rate use of biologic therapies
HBcAb+	175	42.66 ± 12.8	622.19	1.14 (2/175)	1.14 (2/175)	0.32	1.4 (1-2.32)	ETA: 53.33% (176/330)
Chronic HBV	40	17.24 ± 9.11	57.46	20 (8/40)	65 (26/40)	13.92	1.05 (1-2)	ADA: 21.52% (71/330)
								UST:13.03% (43/330)
HCV+	97	15.32 ± 10.72	123.83	3.09 (3/97)	12.37 (12/97)	2.42	1.07 (1-3)	INF: 8.18% (27/330)
								EFA: 1.82% (6/330)
								ALE: 0.91% (3/330)
								SEC: 0.91% (3/330)
								GOL: 0.3% (1/330)
Sum	312	30.9 ± 11	803.48	4.17 (13/312)	12.82 (40/312)	1.62	1.24 (1-3)	330

ADA, Adalimumab; *ALE*, alefacept; *EFA*, efalizumab; *ETA*, etanercept; *GOL*, golimumab; *HBcAb*, HBV core antibody; *HBV*, hepatitis B virus; *HCV*, hepatitis C virus; *INF*, infliximab; *SEC*, secukinumab; *UST*, ustekinumab; *PRX*, prophylaxis; *pts*, patients; *SD*, standard deviation.

Table VI. Summary of reports describing viral reactivation during biologic therapies

Study	Viral profile	No pts	Level of evidence	Biologic therapies	Length of treatment,months	Viral PRX	Hepatitis
Navarro et al, 2013[14]	HCV+	1	B	ETA	23	No	No
Chiu et al, 2013[15]	HCV+	1	B	UST	1	No	No
Ventura et al, 2010[48]	HCV+	1	C(s)	ETA	6	No	No
Koskinas et al, 2013[21]	HBcAb+	1	C	UST	16	No	No
Notarnicola et al, 2014[28]*	HBcAb+	1	C	INF	66	No	No
Chiu et al, 2013[15]	Chronic HBV	2	B	UST	5.5	No	No
Fotiadou et al, 2011[16]	Chronic HBV	1	B	INF	10	Yes	No
Cho et al, 2012[20]	Chronic HBV	3	B	ETA	32.3	1/3	No
Kouba et al, 2012[24]	Chronic HBV	1	C	ADA	0.75	No	Yes
Corpas et al, 2012[25]	Chronic HBV	1	C	ETA, ADA	34	Yes	No
Total chronic HBV	Chronic HBV	8		ETA-4, ADA-2, UST-2, INF-1	19.08	3/8	1/8

ADA, Adalimumab; *B*, retrospective study; *C*, case report; *C(s)*, small case series; *ETA*, etanercept; *HBcAb*, HBV core antibody; *HBV*, hepatitis B virus; *HCV*, hepatitis C virus; *INF*, infliximab; *PRX*, prophylaxis; *UST*, ustekinumab.
*Baseline HBV DNA was not available, yet a high replication rate at the end of follow-up was indicative of viral reactivation.

infection, which is defined by the persistence of HBV DNA in the liver tissue or serum without apparent infection.[70] Positivity for only core antibody might indicate a resolved past infection, an occult HBV infection (OBI), a false-positive HBcAb result, or a replicative HBsAg-negative infection.[71-73] OBIs cannot be confirmed because diagnosis requires liver biopsies being positive for HBV DNA, and the performance of a liver biopsy is not justifiable in patients without clinical signs of hepatitis. Patients regarded as having a past but resolved HBV infection (HBsAg negative, HBcAb positive, HBsAb positive) might instead have an infection with a low viral replication rate.[67] Because viral reactivation in this group occurred primarily in patients with hematologic and lymphoproliferative malignancies treated with rituximab,[66,67] these patients may be regarded as having potential OBIs. Therefore, in our systematic review, patients seropositive for core antibody were grouped together, irrespective of their surface antibody status.

In our systematic review, only 2/175 patients positive for core antibody (1.14%) had viral reactivation, for a 0.32% yearly rate. A recent systematic review including 712 rheumatic patients with resolved HBV infection treated with biologic therapies reported a reactivation rate of 1.7%.[62] The lower calculated risk in our review might reflect the less frequent use of concomitant immunosuppressants in psoriasis than in rheumatic diseases.[13]

The European Association for the Study of the Liver[8] defines chronic HBV as surface antigen positivity, regardless of the status of other serologic markers. Patients with chronic HBV infection are, by far, at greatest risk for HBV reactivation,[74] with reported rates of up to 50% in those undergoing cytotoxic chemotherapy for malignancies without antiviral prophylaxis.[75] Our systematic review

included 40 chronic HBV patients, of whom 8 (20%) experienced viral reactivation, for a yearly reactivation rate of 13.92%.

In immunosuppressed patients with chronic HBV infection, oral anti-HBV agents have proven to be effective in reducing acute exacerbations and preventing fatal complications.[8,76] Accordingly, only 3/26 patients in the systematic review with chronic HBV receiving antiviral prophylaxis experienced viral reactivation compared with 5/14 patients with chronic HBV without prophylaxis, yielding yearly reactivation rates of 7.74% and 26.31%, respectively (Table VI and Supplemental Table II; available at http://www.jaad.org). Therefore, we suggest that all patients with chronic HBV be treated with antiviral agents before biologics are initiated. This is supported by the 2015 European guidelines for the use of anti-TNF therapy for psoriasis, wherein HBV infection is not considered a barrier to immunosuppression therapy.[77]

Chronic HCV infection might be promoted by the upregulation of certain inflammatory cytokines, such as TNF-α.[78] Studies have shown that TNF-α—inhibitor therapy is associated with a drop in viral load and, in some cases, even lead to viral clearance.[14,79] In our analysis, only 3/97 patients with HCV infection (3.09%) had evidence of viral reactivation, with a yearly rate of 2.42%. Considering that 3%-10% of patients with HCV infection show spontaneous fluctuations in viral load of >1 log HCV RNA,[80] our systematic review suggests that biologic therapies pose no risk of HCV reactivation. Nevertheless, because hepatocellular carcinoma[81] and viral reactivation are part of the natural history of HCV, patients should be followed periodically according to the European Association for the Study of the Liver guidelines.[82]

Our analysis has several limitations. First, different biologics, including various agents within the anti—TNF-α family, were shown to confer disparate risk for serious infections among psoriasis patients.[83] Nonetheless, we preemptively decided to pool data from heterogeneous studies evaluating different biologics to make possible an analysis of viral reactivation (which is an uncommon event) among a relatively small number of patients. Moreover, as the average follow-up period was limited, large-scale prospective trials evaluating distinct biologic therapies are in need to validate our conclusions. Second, we included case reports, which are known to have a positive-outcome publication bias.[84,85] The case reports and small case series accounted for only 14.42% (45/312) of the total patients but 41.7% (5/12) of the patients with viral reactivation. Nonetheless, the possible overestimation of reactivation rates strengthens our conclusions regarding the safety profile of biologic therapies.

In conclusion, our study indicates that low-risk patients without hepatitis who are seropositive for HBV core antibody only or HCV infection have minimal risk or no additive risk, respectively, for viral reactivation. Consequently, antiviral prophylaxis seems unnecessary.

In contrast, chronic carriers of HBV have about a 14% risk for yearly viral reactivation. These patients need to be carefully evaluated for the risk:benefit ratio of biologic therapy and receive antiviral prophylaxis.

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APPENDIX I
SEARCH CRITERIA

The following keywords were used for the Pubmed search (viral hepatitis [Title/Abstract] OR HBV [Title/Abstract] OR HCV [Title/Abstract] OR HBsAg [Title/Abstract] OR HBcAb [Title/Abstract] OR surface [Title/Abstract] OR core [Title/Abstract] OR hepatitis B virus [Title/Abstract] OR hepatitis C virus [Title/Abstract] OR hepatitis B [Title/Abstract] OR hepatitis C [Title/Abstract]]] AND (psoriasis [Title/Abstract] OR psoriatic [Title/Abstract]]] AND (etanercept [Title/Abstract] OR adalimumab [Title/Abstract] OR ustekinumab [Title/Abstract] OR infliximab [Title/Abstract] OR cosentyx [Title/Abstract] OR golimumab [Title/Abstract] OR alefacept [Title/Abstract] OR antitumor [Title/Abstract] OR anti TNF [Title/Abstract] OR biologic [Title/Abstract] OR anti-TNFα OR anti-tumour necrosis [Title/Abstract] OR tumour necrosis [Title/Abstract] OR Immunomodulator [Title/Abstract] OR immunosuppressive [Title/Abstract])

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Supplemental Table I. Studies and case reports on treatment with biologic therapies in psoriatic patients with past or chronic HBV and HCV[13-60]

Study	Level of evidence	Pts, n	Virus and mean follow-up, months	Patient subdivision and antiviral therapy	Biologic therapies	Reactivation
Sanz-Bueno et al, 2015[13]	B	20	HBV 40	Isolated 7; no PRX Past 13; no PRX	ADA 13; ETA 14; INF 7; UST 6; EFA 1; mean per patient: 2.05	Isolated 0/7 Past 0/13
Navarro et al, 2013[14]	B	16	HCV 19.82 HBV 25	HCV 11; 2/11 PRX Chronic 5; 5/5 PRX	ETA 11; ADA 3; UST 4; INF 1; mean per patient: 1.19	HCV 1/11 (ETA, 23 M; no PRX, no hepatitis)[*] Chronic 0/5
Chiu et al, 2013[15]	B	18	HCV 9.5 HBV 10.4	HCV 4; no PRX Chronic 11; 4/11 PRX Isolated 1; no PRX Past 2; no PRX	UST 18; mean per patient: 1	HCV 1/4 (1 M, no hepatitis) Chronic 2/11 (5.5 M; no PRX, no hepatitis) Isolated 0/1 Past 0/2
Fotiadou et al, 2011[16]	B	7	HBV 12.14	Chronic 7; 7/7 PRX	ADA 3; ETA 3; INF 1; mean per patient: 1	Chronic 1/7 (INF, 10 M; PRX, no hepatitis)
Morisco et al, 2014[17]	B	46	HCV 46[†] HBV 46[†]	HCV 6; no PRX Isolated 18; no PRX Past 22; no PRX	ETA; ADA; UST; INF; GOL; mean per patient: N/A	HCV 0/6 Isolated 0/18 Past 0/22
Cassano et al, 2011[18]	B	62	~HBV 48	Isolated + past 62; 1/62 PRX	ETA 44; INF 8; ADA 10; mean per patient: 1	Isolated + past 0/62
Prignano et al, 2011[19]	B	17	HCV 11 Isolated + past 7.3 Chronic 10	HCV 5; no PRX Past 11; no PRX Chronic 1[‡]; no PRX	ADA 3; ETA 15: mean per patient: 1.06	HCV 0/5 Chronic 0/1 Past 0/11
Cho et al, 2012[20]	B	7	HBV 28.9	Chronic 7; 2/7 PRX	ADA 1; ETA 6; mean per patient: 1	Chronic 3/7 (ETN, 32.3 M; 1/3 PRX, no hepatitis)
Koskinas et al, 2013[21]	C	1	HBV 16	Past 1; no PRX	UST 1; mean per patient: 1	Past 1/1 (16 M; no hepatitis)
Conde-Taboada et al, 2009[22]	C	1	HBV 15	Chronic 1; PRX	INF 1; mean per patient: 1	Chronic 0/1
Nosotti et al, 2010[23]	B	7	Isolated + past 8.75 Chronic 12.13	Chronic 1; PRX Isolated + past 6; no PRX	ADA 3; ETA 4; mean per patient:1	Chronic 0/1 Past 0/6
Kouba et al, 2012[24]	C	1	HBV 0.75	Chronic 1; no PRX	ADA 1; mean per patient: 1	Chronic 1/1 (0.75 M; no PRX, hepatitis)
Corpas et al, 2012[25]	C	1	HBV 34	Chronic 1; PRX	ETA 1; ADA 1; mean per patient: 2	Chronic 1/1 (34 M; PRX LAM + TEN, hepatitis)
Salvi et al, 2016[26]	C	1	HCV 24	HCV 1; PRX	ETA 1; ADA 1; INF 1; mean per patient: 3	HCV 0/1
Raymundo et al, 2016[27]	C	1	HBV 39	Chronic 1; PRX	UST 1; mean per patient: 1	Chronic 0/1
Notarnicola et al, 2013[28]	C	1	HBV 66	Past 1; no PRX	INF 1; mean per patient: 1	Past 1/1 (no PRX, no hepatitis)[§]

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Abuchar et al, 2013[29]	C	1	HCV 7	HCV 1; no PRX	ETA 1; UST 1; mean per patient: 2	HCV 0/1
Babino et al, 2013[30]	C	1	HCV 11.2	HCV 1; PRX	ETA 1; mean per patient: 1	HCV 0/1
Prignano et al, 2011[31]	C	1	HBV 5.6	Chronic 1; PRX: LAM	ETA 1; mean per patient: 1	HBV 0/1
Zanni et al, 2011[32]	C	3	HCV 6	HCV 3; no PRX	ETA 3; mean per patient: 1	HCV 0/3 1 patient had a rise of hepatic enzymes
Mederacke et al, 2011[33]‡	C	1	HBV 14	Chronic + HCV 1; PRX: IFN-α + RIB	ETA 1; mean per patient: 1	Chronic 0/1
Gandhi et al, 2010[34]	C	1	HCV 12	HCV 1; no PRX	ETA 1; mean per patient: 1	Chronic 0/1
De Simone et al, 2006[35]	C	2	HCV 12	HCV 2; no PRX	ETA 2; mean per patient: 1	HCV 0/2
Magliocco & Gottlieb, 2004[36]	C	3	HCV 5	HCV 3; 1/3 PRX	ETA 3; mean per patient: 1	HCV 0/3
Cecchi & Bartoli, 2006[37]	C	1	HCV 12	HCV 1; no PRX	ETA 1; mean per patient: 1	HCV 0/1
Rokhsar et al, 2006[38]	C	1	HCV 12	HCV 1; no PRX	ETA 1; mean per patient: 1	HCV 0/1
Collazo et al, 2008[39]	C	1	HCV 3	HCV 1; no PRX	ETA 1; mean per patient: 1	HCV 0/1
Behnam et al, 2010[40]	C	1	HCV 5.6	HCV 1; PRX: IFN-α + RIB	ETA 1; mean per patient: 1	HCV 0/1
Garavaglia & Altomare, 2010[41]	B	5	HCV 24	HCV 5; 1/5 PRX	ETA 5; mean per patient: 1	HCV 0/5‖
Paradisi et al, 2010[42]	A	2	HCV 12	HCV 2; no PRX	ETA 2; mean per patient: 1	HCV 0/2
Cassano & Vena, 2008[43]	C	1	HCV 5.6	HCV 1; no PRX	ETA 1; mean per patient: 1	HCV 0/1
Alcaide et al, 2008[44]	C	1	HCV 5.6	HCV 1; no PRX	ETA 1; mean per patient: 1	HCV 0/1
Thaçi et al, 2005[45]	C	2	HCV 5.6	HCV 2; no PRX	ALE 1; mean per patient: 1	HCV 0/2
Malara et al, 2012[46]	C	1	HCV 6.5	HCV 1; PRX NA	UST 1; mean per patient: 1	HCV 0/1
Prignano et al, 2009[47]	C	1	HCV 11	HCV 1; no PRX	ETA 1; ALE 1; mean per patient: 2	HCV 0/1
Ventura et al, 2010[48]	C	2	HCV 12	HCV 2; no PRX	ETA 2; mean per patient: 1	HCV 1/2 (6 M, no hepatitis)
Aslanidis et al, 2007[49]	C	1	HCV 12	HCV 1; PRX	INF 1; ADA 1; mean per patient: 2	HCV 0/1
Di Lernia & Guareschi, 2010[50]	C	1	HCV 10	HCV 1; PRX NA	INF 1; mean per patient: 1	HCV 0/1
Di Lernia et al, 2013[51]	C	1	HCV 30.8	HCV 1; PRX	ETA 1; mean per patient: 1	HCV 0/1
Linardaki et al, 2007[52]	C	1	HCV 24	HCV 1; PRX	ETA 1; mean per patient: 1	HCV 0/1
Costa et al, 2014[53]	B	15	HCV 12	HCV 15; no PRX	ETA 8; ADA, 7; mean per patient: 1	HCV 0/15
Laurenti et al, 2013[54]	B	8	Isolated + past 44.57¶ Chronic 24	Chronic 1; PRX Isolated + past 7; no PRX	ADA 8; mean per patient: 1	Chronic 0/1 Isolated 0/7
Bartalesi et al, 2013[55]	C	1	HCV 11.2	HCV 1; PRX	ETA 1; mean per patient: 1	HCV 0/1
Cavazzana et al, 2008[56]	B#	1	HCV 14#	HCV 1; no PRX	ETA 1; mean per patient: 1	HCV 0/1
Chiricozzi et al, 2012[57]	C	1	HCV 25.5	HCV 1; no PRX	INF 1; ADA 1; ETA 1; mean per patient: 3	HCV 0/1
Gisondi et al, 2009[58]	C	5	HCV 10.8	HCV 5; no PRX	EFA 5; mean per patient: 1	HCV 0/5
Esposito et al, 2007[59]	B	8	HCV 5.6	HCV 8; PRX N/A	ETA 8; mean per patient: 1	HCV 0/8

Continued

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Supplemental Table I. Cont'd

Study	Level of evidence	Pts, n	Virus and mean follow-up, months	Patient subdivision and antiviral therapy	Biologic therapies	Reactivation
Di Nuzzo et al, 2016[60]**	C	2	HCV 42.5	HCV 2; 1/2 PRX	ETA 2; mean per patient: 1	HCV 0/2
Present study	B	28	HCV 17.64 Chronic 12 Isolated + past 53	HCV 3; no PRX Chronic 1; PRX Isolated + past 24; no PRX	ETA 26; ADA 15; UST 11; INF 4; SEC 3; GOL 1; ALE 1; mean per patient: HBV 2.32; HCV 1	HCV 0/3 Chronic 0/1 Isolated 0/7 Past 0/17

A, Prospective controlled study; *ADA*, adalimumab; *ALE*, alefacept; *B*, retrospective study; *C*, case report or case series; *EFA*, efalizumab; *ETA*, etaneracept; *GOL*, golimumab; *HBV*, hepatitis B virus; *HCC*, hepatocellular carcinoma; *HCV*, hepatitis C virus; *INF*, infliximab; *LAM*, lamivudine; *M*, months; *N/A*, not available; *PRX*, prophylaxis; *RIB*, ribavirin; *SEC*, secukinumab; *TEN*, tenofovir; *UST*, ustekinumab.

*Another patient was suspected of reactivation with doubling of hepatic enzymes, but viral load was not available.

†Part of this follow-up period (length not stated in text) was with nonbiologic therapies.

‡Counted once only as chronic HBV, although concomitantly infected with HCV.

§Baseline HBV DNA was not available, yet a high replication rate at the end of follow-up was indicative of viral reactivation.

‖One patient experienced a moderate rise in viral load of less than 1 log 10, requiring the start of an antiviral treatment.

¶Estimated follow-up period ranging between year of starting adalimumab and 2012 (1 year before data of publication).

#This study included 6 patients with chronic rheumatic diseases for a median treatment period of 14 months. One patient had psoriatic arthritis.

**Both patients developed HCC.

Supplemental Table II. Total and yearly viral reactivation rates in chronic HBV carriers with and without antiviral treatment

Viral prophylaxis status	Pts, n	Follow-up duration mean ± SD, months	Combined follow-up duration, years	Reactivation rate, % (n/total)	Yearly reactivation rate, %
Antiviral treatment	26	17.89 ± 8.63	38.76	11.54 (3/26)	7.74
No prophylaxis	14	16.29 ± 9.71	19	35.71 (5/14)	26.31

Pts, Patients; *SD*, standard deviation.